MICROECONOMICS

Central Problems of an Economy

- **What** goods and services shall be produced in an economy and **how much**?
- **How** shall the goods and services be produced?
- **For Whom** the goods and services are to be produced?

Human needs are **unlimited,** while the resources are **limited**. Thus, there exists scarcity of resources in an economy. Better resource allocation is called for in this scenario so as to resolve the Central Problems in an Economy.

Central Problems of an Economy

Economic problems are solved through:

1.Price Mechanism in a Market Economy

2.State Machinery of Central Planning in a Socialist Economy

3.Price Mechanism with active intervention and regulation of the state in a Mixed Economy

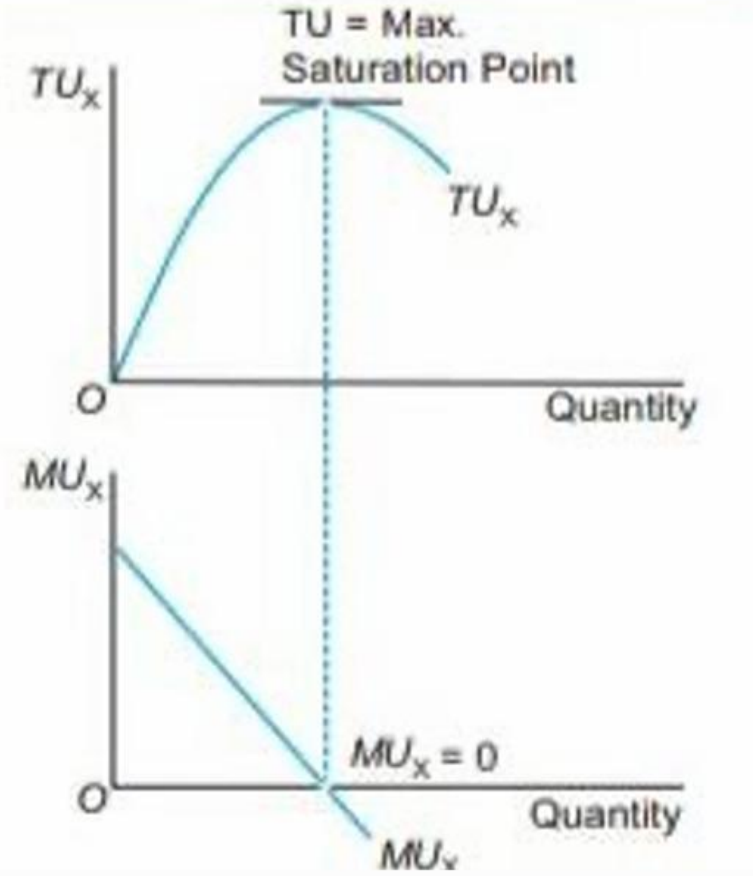
UTILITY

- **Utility:** The ability of a good to satisfy a want. "Util" is the arbitrary unit of measurement of utility.
- **Total Utility (TU) :** The total satisfaction received from consuming a good or a service.
- **Marginal Utility (MU) :** The extra utility derived from consuming an additional unit of a good or a service.

Quantity	TU_x	MU_x
0	0	-
1	10	10
2	16	6
3	20	4
4	22	2
5	22	0
6	20	-2

UTILITY

- TU increases at an increasing rate, then increases at a decreasing rate, reaches maximum and then falls.
- When TU is rising, MU is negative and falling.
- When TU reaches its peak, MU becomes zero.
- When TU falls, MU becomes negative.
- MU is the first derivative of TU.

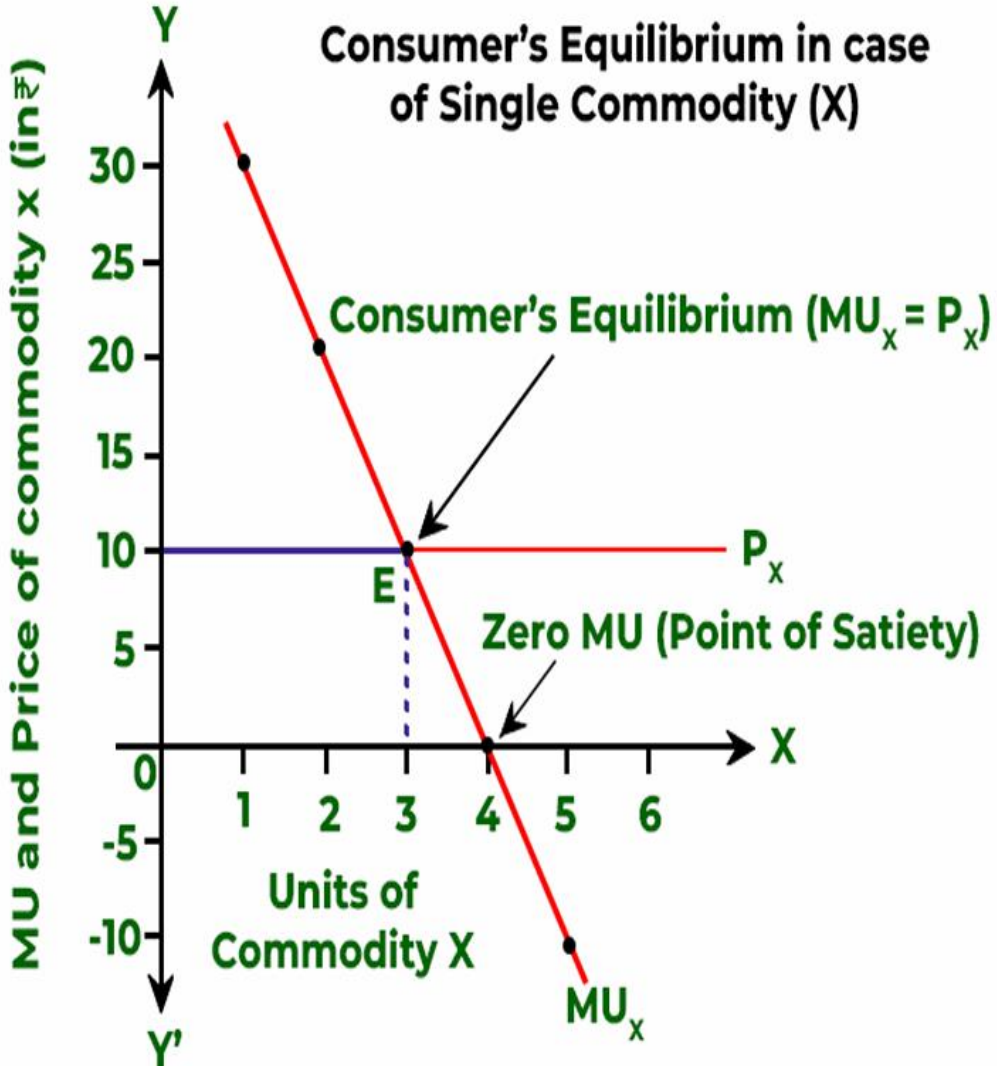


Consumer's Equilibrium : One-Commodity Case

- Price is given and constant (P_x)
- Consumer's equilibrium is attained at the point "E" where the MU_x curve intersects the price line (P_x).

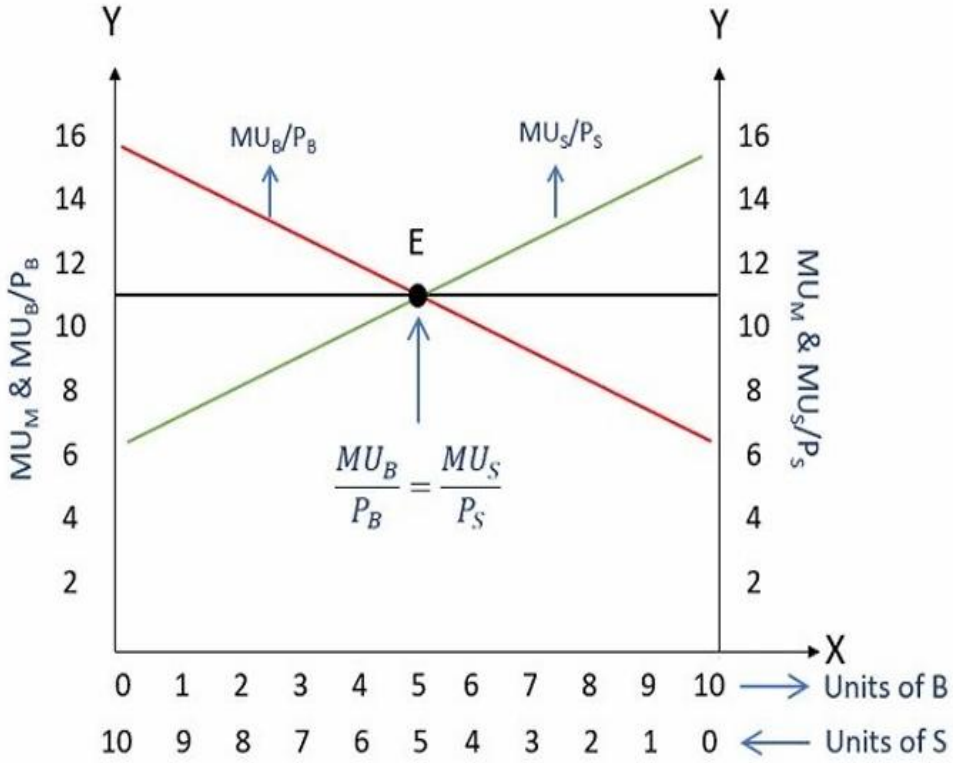
At this point, the price of good X equals the MU (Marginal Utility) derived from it.

Consumer's Equilibrium: $MU_x = P_x$



Consumer's Equilibrium : Two-Commodity Case

- In the figure, the consumer allocates his income between two goods; good B and good S in such a way that the last rupee spent on each good gives the same amount of satisfaction.
- Marginal Utility of Money (MU_m) is assumed to be constant.
- Equilibrium is reached at point E where, $\frac{MU_b}{P_b} = \frac{MU_s}{P_s} = MU_m$
- Any deviation from the equilibrium will force back a reversal.
- If $\frac{MU_b}{P_b} > \frac{MU_s}{P_s}$, the consumer will consume more of good B as it gives more utility and eventually the equilibrium point is reached.
- If $\frac{MU_s}{P_s} > \frac{MU_b}{P_b}$, the consumer will consume more of good S as it gives more utility and moves leftwards on x-axis to reach the equilibrium.



Consumer's Equilibrium : Two-Commodity Case

Units	MU_x	MU_y	P_x	P_y	$\frac{MU_x}{P_x}$	$\frac{MU_y}{P_y}$
1	40	55	4	5	10	11
2	36	50	4	5	9	10
3	32	30	4	5	8	6
4	28	20	4	5	7	4
5	24	15	4	5	6	3
6	20	10	4	5	5	2
7	12	5	4	5	3	1

Equilibrium is attained when 7 units of X and 5 units of Y is consumed and the total expenditure is INR 53. If the consumer's income is less than 53, he will not be able to consume this bundle.

Consumer's Equilibrium : Ordinal Approach

- **A consumer attains equilibrium when the budget line is tangent to the highest possible indifference curve. At this point E, the consumer consumes the bundle which is affordable (given by the budget line) and gives the maximum satisfaction (on highest IC).** Though points M and N lie on the same budget line, they do not yield the same satisfaction as point E.
- At equilibrium point E, slope of Indifference curve equals the slope of the budget line.

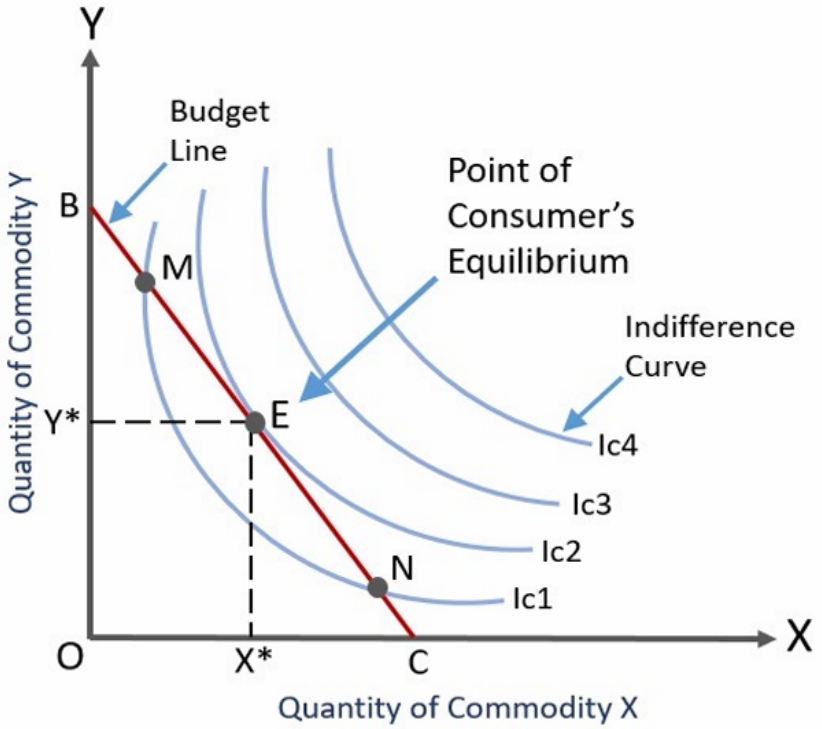
$$MRS_{xy} = \frac{P_x}{P_y}$$

- An Indifference Curve (IC) is the locus of all points that shows different combinations of two goods that yield the same level of satisfaction to the consumer. The ICs are negatively sloped and convex to the origin. No two indifference curve intersect each other. **(WHY?)**
- Marginal Rate of Substitution (MRS) is the slope of an IC curve and shows the amount of a good that the consumer is willing to give up for an additional unit of another good while remaining on the same indifference curve. $MRS_{xy} = \frac{\Delta Y}{\Delta X}$. **Why is MRS diminishing throughout the IC?**

The limitation on the goods and services that a consumer can buy, imposed by his/her limited income and prices of goods is the budget constraint. A budget line shows various combinations of two goods that a consumer can buy with the same level of income. Equation of a budget line:

$$M = P_x.X + P_y.Y$$

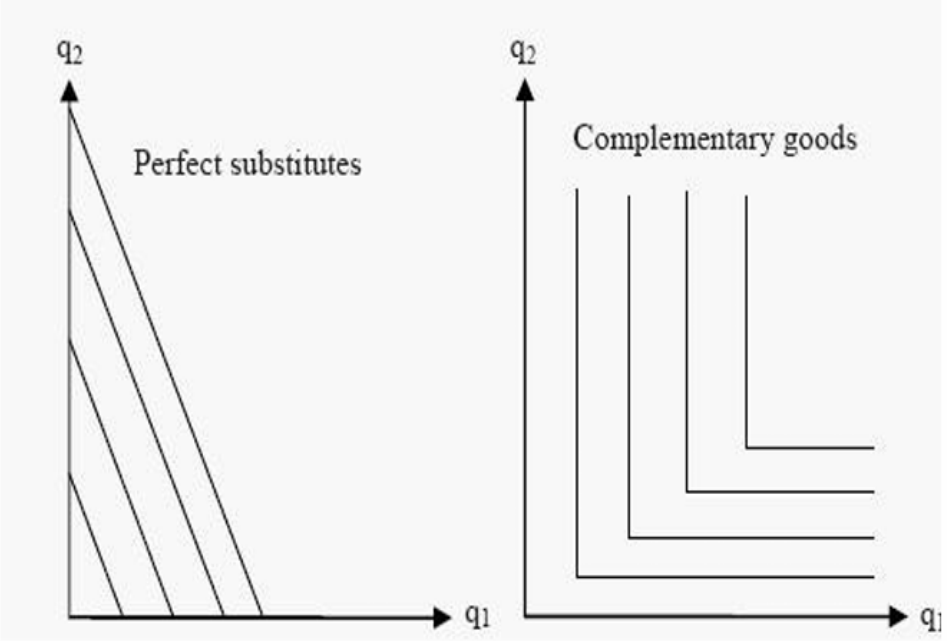
- Slope of a budget line is the ratio of prices of the goods $(\frac{P_x}{P_y})$



IC Curves and Budget Lines

- Incase of substitute goods, the indifference curves will be downward sloping linear curves with constant MRS.
- If the goods are complementary goods, the ICs will be L-shaped (Why?).

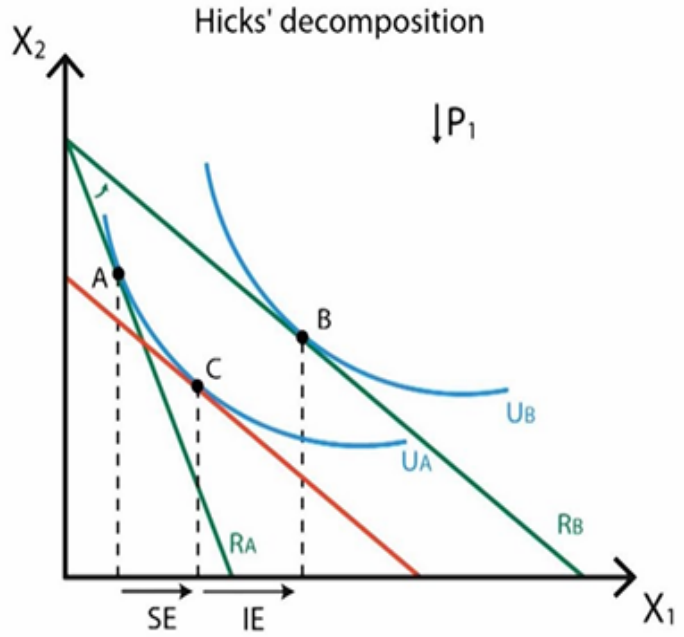
- If there is a rise in income, the budget line shifts rightward and vice-versa.
- If there is fall in the price of one good, then the budget line shifts outward on the respective axis and if there is a rise in the price of one good, the budget line shifts inward on the respective good's axis (Why?).



Price Effect : Hicksian Decomposition

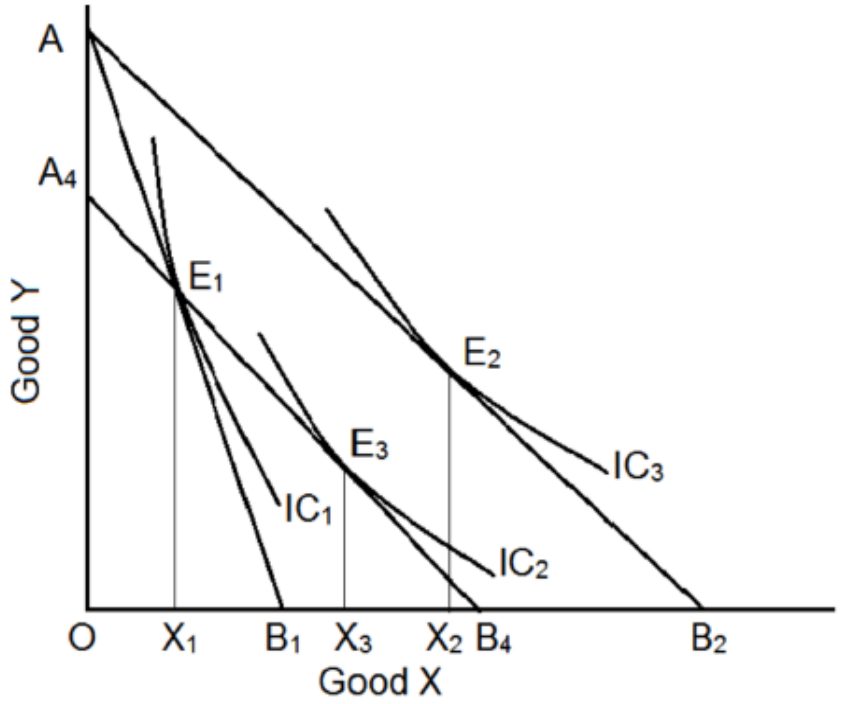
The change in quantity demanded of a commodity owing to the change in its price is called the Price Effect. This Price Effect can be decomposed to Income Effect and Substitution Effect.

The change in quantity consumed of a commodity owing to the commodity becoming relatively cheaper/expensive due to a fall/rise in its price is the Substitution Effect. The change in quantity consumed of a commodity due to a fall/rise in the real income of a consumer owing to the increase/decrease of price of the commodity is the Income Effect.



Price Effect : Slutsky Decomposition

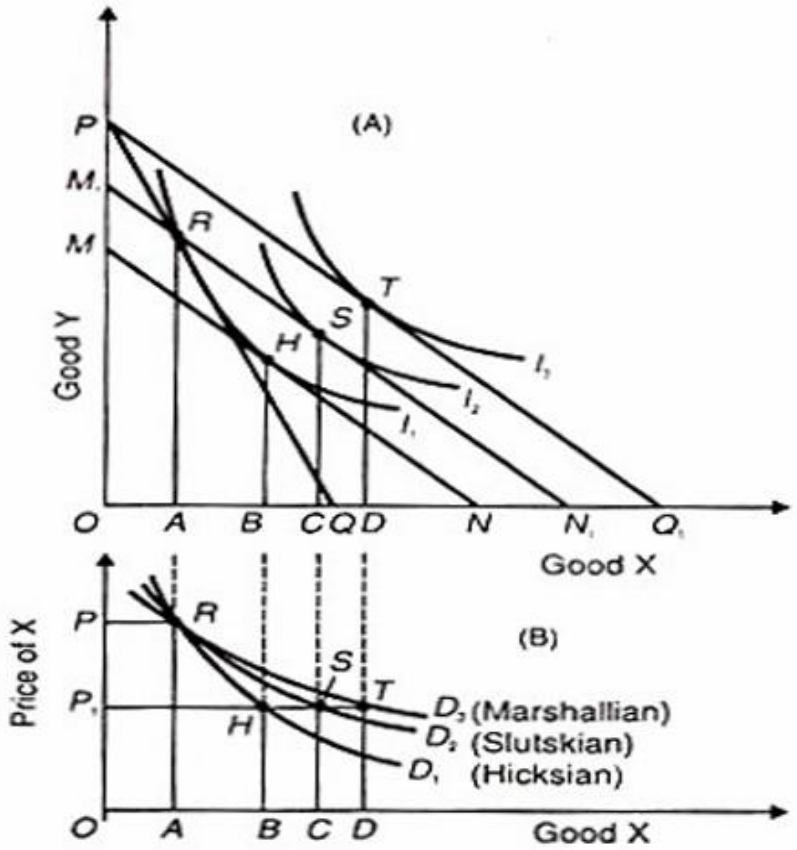
- The compensated budge line A_4B_4 in this case is drawn in such a way that the extra money income that accrue to the consumer owing to a fall in the price of X is taken away and the consumer consumes the previous bundle as if there was no change in income.
- X_1 to X_3 is the Substitution Effect and X_3 to X_2 is the Income Effect.



Derivation of Demand Curve

The Price Consumption Curve is the locus of points that shows the equilibrium quantities of two goods at different price levels. The PCC indicates the change in quantity demanded of a commodity owing to a change in its price. (Panel A)

The demand curve of a good is derived by plotting the prices and corresponding quantity demanded. (Panel B)



THANK YOU